USER EXPERIENCE

The adage goes "the first impression is the last impression", and VR experiences at home typically don't give off a great first impression. In an article published on Bootcamp, UX practitioner Larry Pelty recounted his first experience with a virtual reality headset. He said, "Users who need to create new, or access existing, accounts for applications face difficulties in virtual reality using controllers or hand tracking with limited functionality. Without access to traditional input devices, such as keyboards and mice, or assisting tools, like desktop or mobile password managers, users must find workarounds that typically involve leaving the VR space or removing some or all of the associated equipment. Providing ways of managing or accessing accounts, while maintaining a high level of security and staying within the boundaries of the VR space and its related equipment, will provide for a more comfortable and seamless experience." His entire article drives home the point that while relatively-moderately-priced, standalone VR headsets are here, VR may lose users due to the cumbersome first-time experience.

Another issue that has been known to hamper user experience when using VR equipment is nausea and disorientation. Virtual Reality sickness is an actual term with a whole Wikipedia page dedicated to it. VR sickness is caused when exposure to a virtual environment causes motion sickness-like symptoms. Symptoms can include nausea, vomiting, fatigue, disorientation, headache and general discomfort. Where it differs from motion sickness is that it is induced due to the perception of motion rather than there actually being motion. Studies have found that playing in a seated position can lessen the effects of VR sickness, however, for most interactive VR, playing in a standing position provides more immersiveness. The weight of the headsets is also a deterrent in allowing VR to take off since most headsets are pretty heavy for long periods of usage.

Playing in a seated position can reduce the effects of VR sickness

LACK OF COMPELLING CONTENT

The pandemic ideally should have been VR's moment to shine and breakthrough since millions were locked inside their homes for months together. However, that wasn't really the case. Gartner analyst Tuong Nguyen said, "Hardware updates (to VR) are good, but creating more content and ways to create content, and improving the gadgets' usability, is needed for the tech to fully mature". He continued, "Creating content for VR is a heavy lift. A lot of companies out there are creating platforms you can build stuff on. But the talent and content pipeline isn't there". To create a mass-appealing technology, it is absolutely necessary to have a slew of quality content to go with it. This isn't really the case with VR. A few smash hits like Half-Life: Alyx, Superhot VR, Beat Saber, and Box VR have got a good amount of recognition, however, their sales still aren't anywhere near AAA titles or Nintendo Switch games.

However, headsets such as the Quest 2 can be a turning point for VR. Its lower cost and wireless connectivity abilities are two big advantages. Omdia estimates that over 2.3 million Quest 2s were sold globally just in the last quarter of 2020, which is a step in the right direction. This will hopefully cause more of a stir in the VR development circles, which in turn, should result in more compelling content being created by existing or new companies.

PRIVACY AND SECURITY CONCERNS

As VR technologies are being embraced for applications other than traditional media such as healthcare and education, concerns surrounding privacy and security become more prevalent. The possible risks of breaches involving data theft are a real concern. Another risk is the hacking of recordings and other private information when the products are in use. Sensitive medical and personal data is not something you want getting into the hands of strangers, especially those with malicious intent. VR usually needs user data to operate and most of the information needed is usually highly sensitive and personal; including behavioural data. Body movement tracking data, VR online transactions, medical information, personal details such as addresses, age, gender and more are often what's at stake here. Since VR headsets are usually connected to a PC or at least a mobile app, DDoS attacks and other malware infections can be easy to carry out.

Security protocols need to be implemented at a manufacturer level. The tools used when using VR gear such as microphones, cameras and more need to be encrypted from the ground up. Beyond the built-in security capabilities of a VR headset, companies also need granular control to help make decisions about how the information is shared. Needless to say, security and privacy protocols need to be implemented to have a secure experience when using Virtual Reality technologies. For now, users must ensure to research the VR headset they're thinking about purchasing and reviewing the privacy policies of the headsets and software used. d